

JOINING PROCESSES

ME1031 – Engineering Skill Development

Department of Mechanical Engineering, Sri Lanka
Institute of Information Technology



GIF courtesy: <https://tenor.com/view/weld-welding-soladura-soldar-braze-gif-13004654>

Presented by Dr. L U Subasinghe | PhD (SG), BSc Eng Hons (SL), CEng.MIE(SL), AMIMechE

Senior Lecturer

Department of Mechanical Engineering, University of Moratuwa

lihils@uom.lk

LEARNING OUTCOMES

- At the end of this section, students should be able to
 - **LO-6: Use of basic hand tools and machine tools in engineering workshop for machining processes and joining metals**

OUTLINE

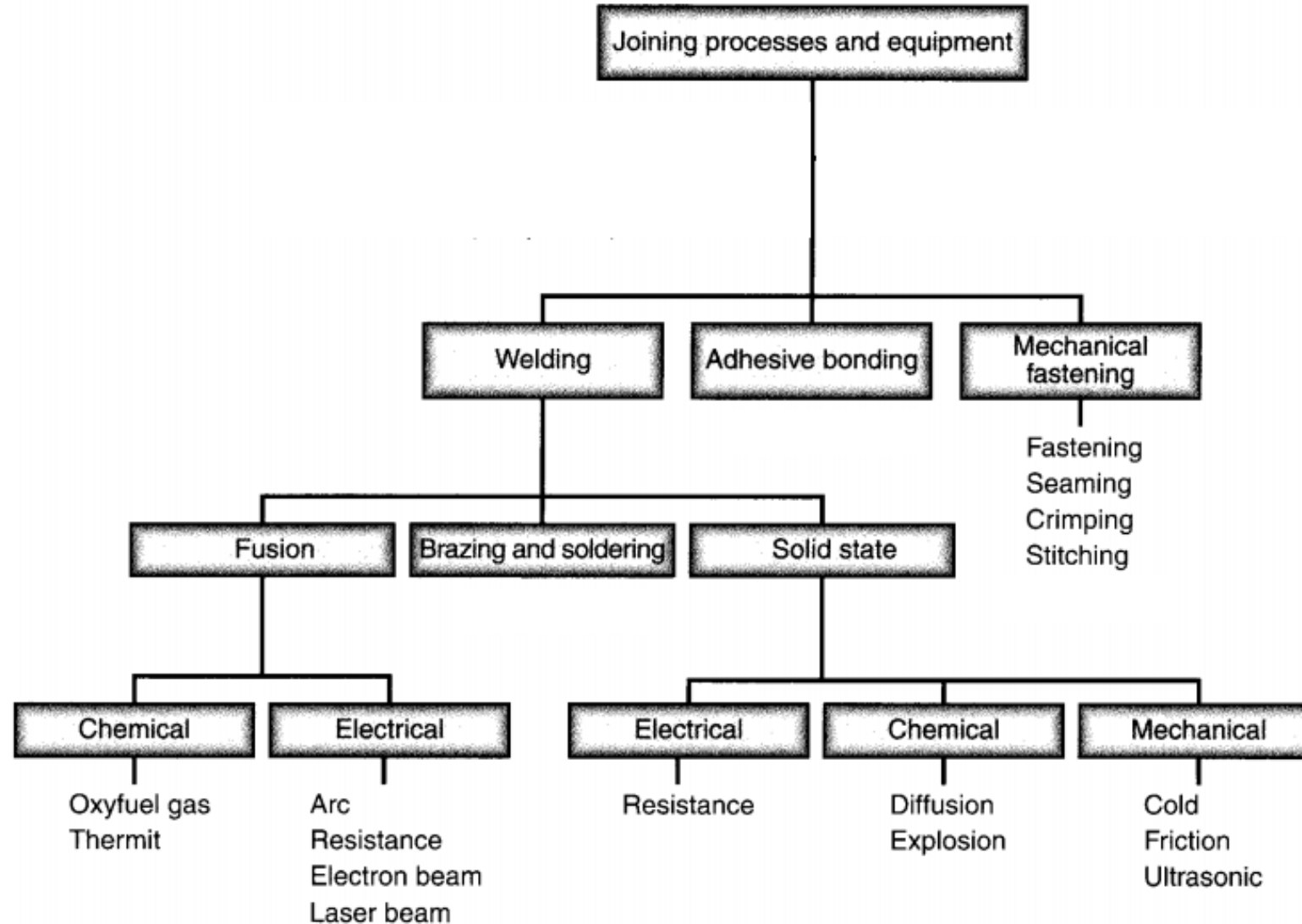
- Introduction to joining processes
 - Welding, adhesive bonding, and mechanical fastening
- Fusion welding
- Solid-state welding
- Brazing and Soldering

INTRODUCTION TO JOINING PROCESSES

IMPORTANCE OF JOINING PROCESSES

- Impossible to manufacture as a single piece
- Easier and more economical to manufacture as separate parts
- Assemblies need to be disassembled for maintenance and replacement purposes
- Different properties may be desirable for functional purposes
- Easier for transportation as small, distinct parts

JOINING PROCESS CLASSIFICATION



SAFETY IN WELDING

- Welders have to protect themselves against spark, hot metal spatter, ultraviolet, infrared and visible light rays, welding fumes, and other hazards



FUSION WELDING

FUSION WELDING PROCESSES

- Fusion welding is *melting* together and *coalescing* materials by means of *heat*.
- **Filler material** - Metals added to the weld area during welding

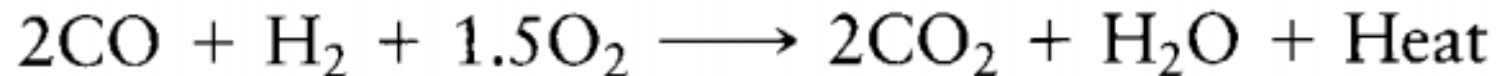


DIFFERENT TYPES OF FUSION WELDING PROCESSES

1. Oxyfuel-gas welding (OFW)
2. Shield metal arc welding (SMAW)
3. Submerged-arc welding (SAW)
4. Gas metal-arc welding (GMAW)
5. Flux-cored arc welding (FCAW)
6. Gas tungsten arc welding (GTAW)
7. Plasma-arc welding (PAW)
8. Thermit welding (TW)
9. Electron-beam welding (EBW)
10. Laser-beam welding (LBW)

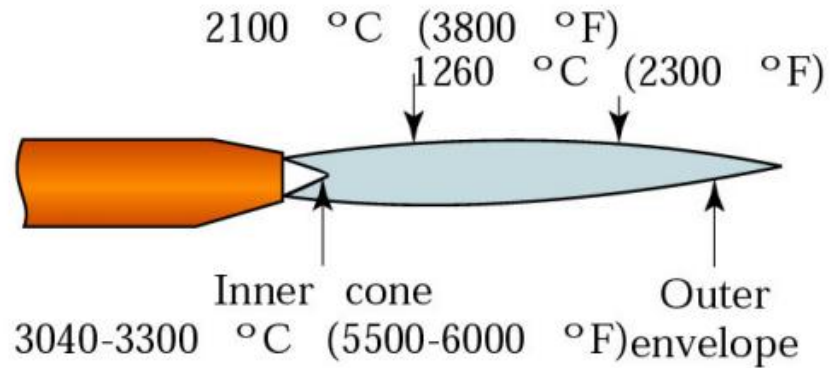
OXYFUEL-GAS WELDING (OFW)

- A welding process that uses a fuel gas combined with oxygen to produce a flame.
- Flame is used to heat and melt the metals at the joint
- Most common fuel → Acetylene → Oxyacetylene-gas welding → OAW

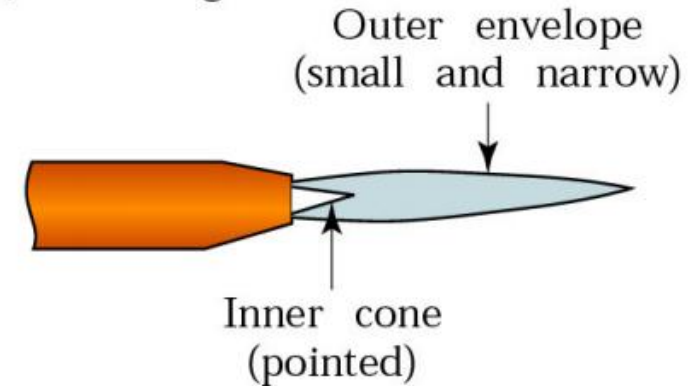


FLAME TYPES

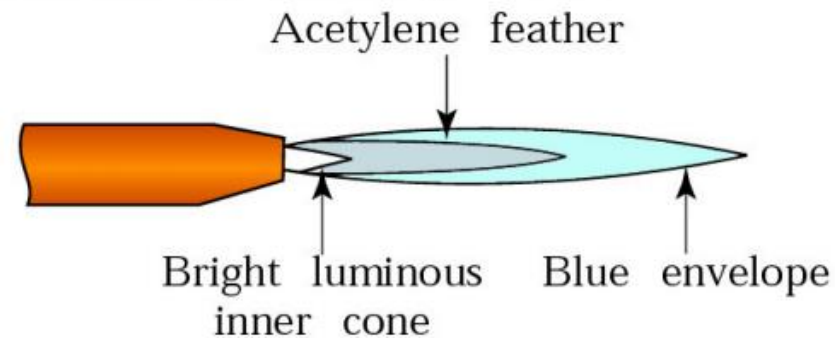
(a) Neutral flame



(b) Oxidizing flame



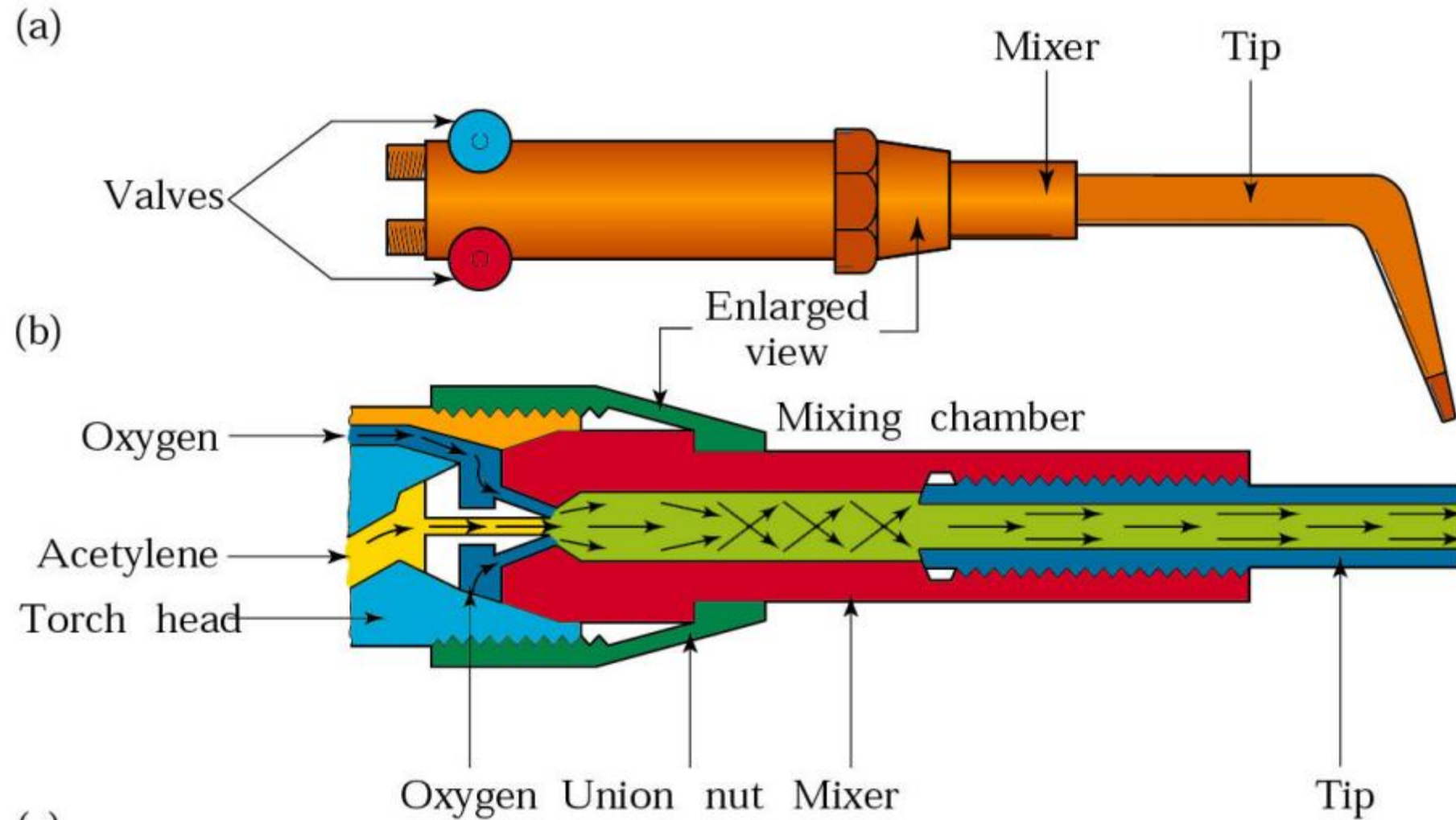
(c) Carburizing (reducing) flame



FLAME TYPES

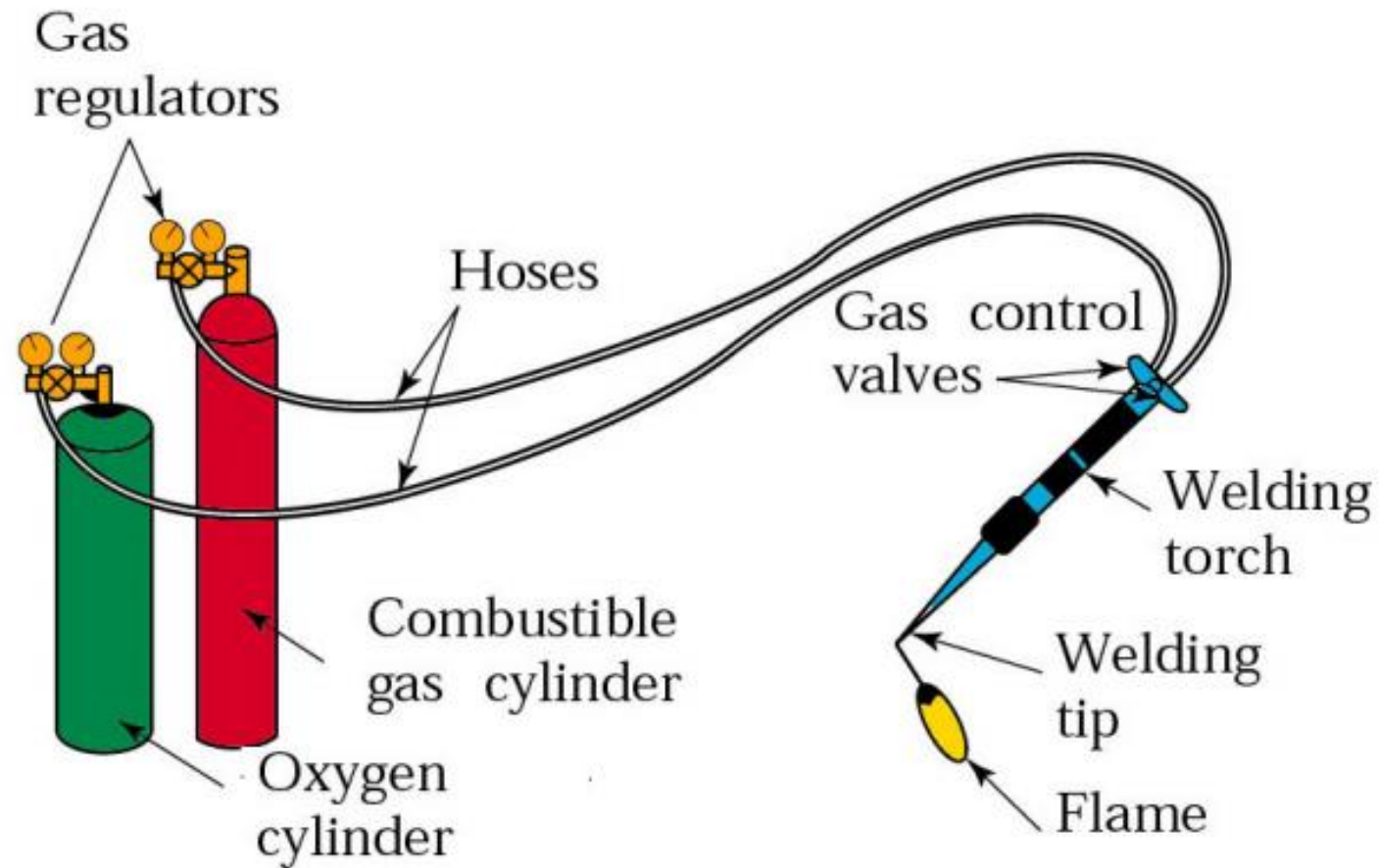
- Neutral flame → No excess oxygen → Stoichiometric combustion
- Oxidizing flame → More oxygen → Oxidize the metal
 - Not good for steels
 - Good for copper and copper based alloys
- Carburizing flame → More acetylene → Low temperature flame
 - Good for brazing, and soldering

OAW EQUIPMENT



OAW EQUIPMENT

- Other equipment



Kalpakjian, S., & Schmid, S. R. (2014). Manufacturing Engineering and Technology (7th ed.). Pearson Education South Asia.
<https://www.youtube.com/watch?v=fs1UhhJH0E8>
<https://www.youtube.com/watch?v=k1hnYforQb4>
<https://www.youtube.com/watch?v=XJ1Loh7eF-A>

ARC WELDING PROCESSES

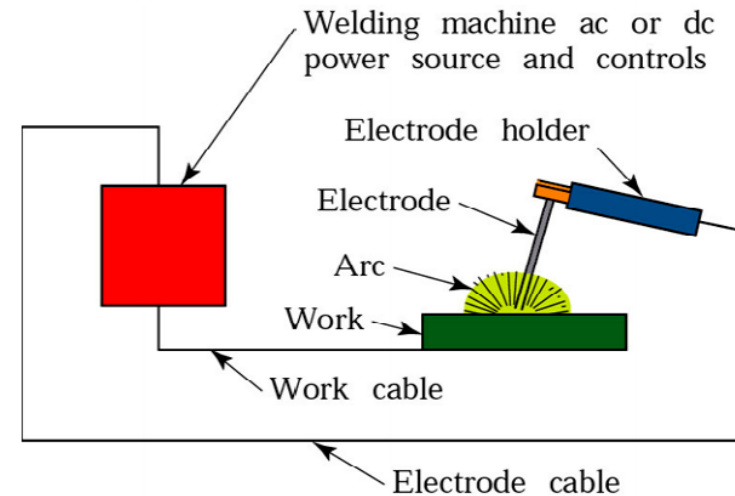
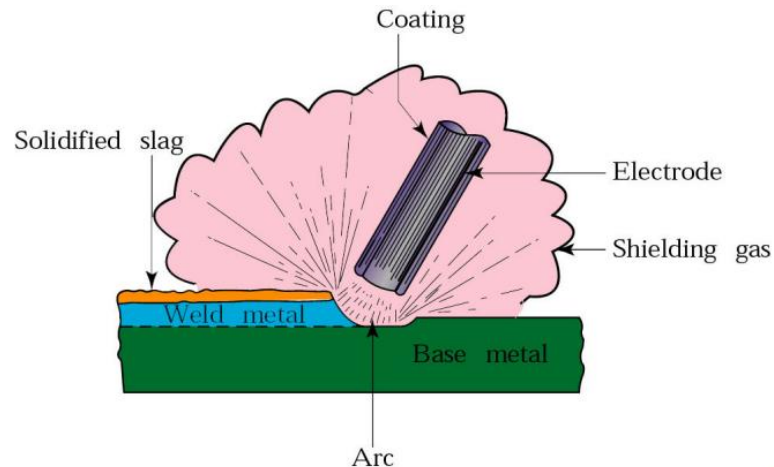
- Heat is produced from electrical energy
- AC or DC power supply produces an arc between the tip of the electrode and the workpiece
- Temperature → 30,000° C
- Two different types
 - Non consumable electrode type
 - Consumable electrode type

SHIELD METAL ARC WELDING (SMAW)

- Most common method used in the industry
- Consumable electrode
- The electric arc is generated by touching the tip of a coated electrode against the workpiece and withdrawing it quickly to a distance sufficient to maintain the arc
- Electrode is a thin, long rod → AKA stick welding

SHIELD METAL ARC WELDING (SMAW)

- Heat generated melts a portion of the electrode tip, its coating, and the base metal in the immediate arc area
- Electrode metal and coating substances are mixed with the base metal



SHIELD METAL ARC WELDING (SMAW)

- Electrode coating deoxidizes the weld area and provides shielding gas to protect it from oxygen in the environment
- Simple, versatile and cheap
- Equipment → Power supply, cables, and electrode holder only
- Slag needs to be removed after each weld bead → Weld chipping, wire brushing
- Applications
 - General construction, ship building, pipelines

Kalpakjian, S., & Schmid, S. R. (2014). Manufacturing Engineering and Technology (7th ed.). Pearson Education South Asia.

<https://www.youtube.com/watch?v=elmDvqdeMKI>

<https://www.youtube.com/watch?v=I1P5GuVHFQA>

<https://www.youtube.com/watch?v=pqGI0iM0dyU>

FILLER METALS

- Supply additional metals to the weld zone
- Available as filler rods or wires
- May be bare or coated with *flux*
- Usage of flux
 - Retard oxidation of the surface by producing a gaseous shield
 - Dissolve and remove oxides and other impurities from the weld zone
 - Developed *Slag* (oxides, fluxes and filler rod-coating materials) protects the weld pool against oxidation when it cools

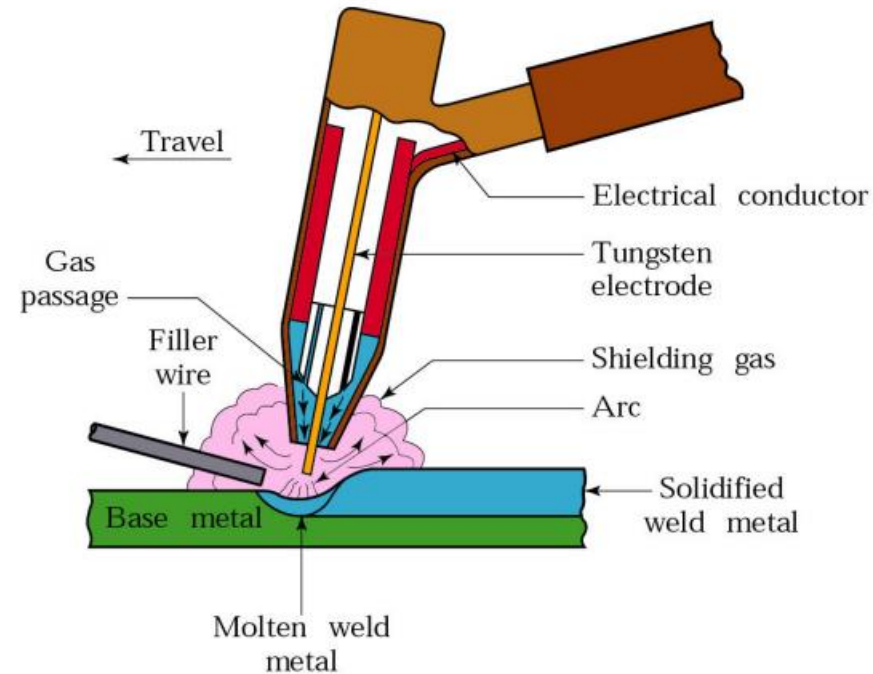
ELECTRODE NOMENCLATURE

- EXXX
- E stands for arc welding electrode
- First two digits x 1000 = Minimum tensile stress in psi
- 3rd digit → Position
 - 1 = All positions
 - 2 = Flat position or horizontal fillets
- 4th digit → type of coating and current

GAS TUNGSTEN ARC WELDING (GTAW)

- Commonly known as TIG (tungsten inert gas) welding
- Non consumable tungsten electrode
- Filler metal is used but without flux
- Shielding gas → Argon or Helium (or mixture of both)
- Good for thin-metals
- Al, Mg, Ti

GAS TUNGSTEN ARC WELDING (GTAW)



- Expensive because of the cost of inert gas
- Equipment is portable
- Very high quality and surface finish

SOLID-STATE WELDING

SOLID-STATE WELDING

- Joining takes place without fusion at the interface of the two parts
- No liquid or molten phase in the joint
- Can be done with or without applying heat
- Application of pressure is essential
- Plastic deformation is involved in mating process

DIFFERENT TYPES OF SOLID-STATE WELDING PROCESSES

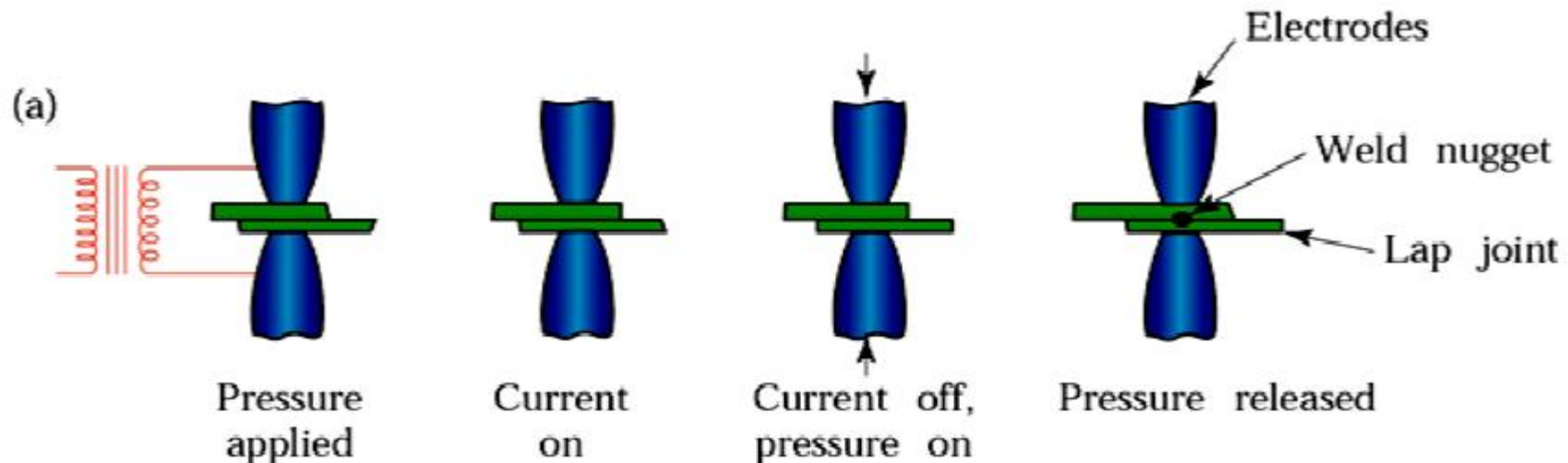
1. Cold welding (CW)
2. Friction welding (FW)
3. Resistance welding (RW)
 - i. Resistance spot welding (RSW)
 - ii. Resistance seam welding (RSEW)
4. Flash welding (FW)
5. Explosion welding (EXW)

RESISTANCE WELDING (RW)

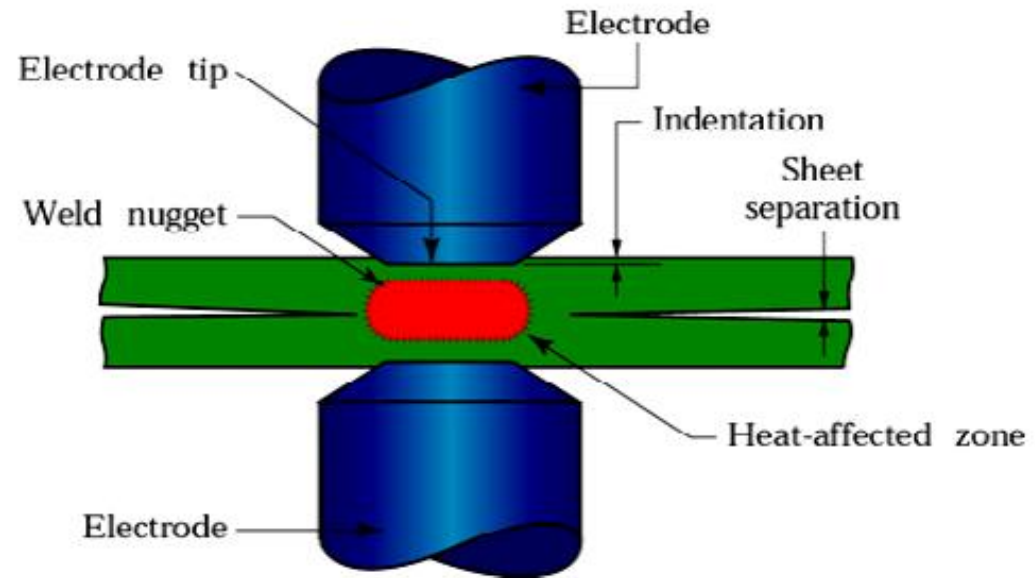
- Heat is produced by means of electrical resistance across the two parts to be joined
- No need of consumable electrodes, shielding gases or flux
- Mating surfaces should be clean → No oil, paint or oxides
- Very simple, less expensive
- No need of skilled workers
- Similar or dissimilar metals can be joined

RESISTANCE SPOT WELDING (RSW)

- The tips of two opposing solid, cylindrical electrodes touch a lap joint of two sheet metals
- Resistance heating produces a spot weld
- Pressure is applied until the current is turned off and the joint has solidified

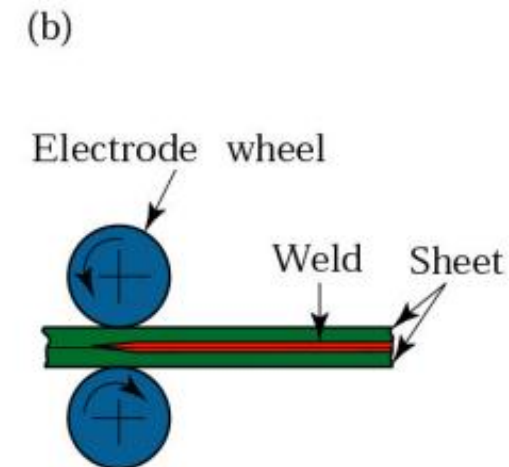
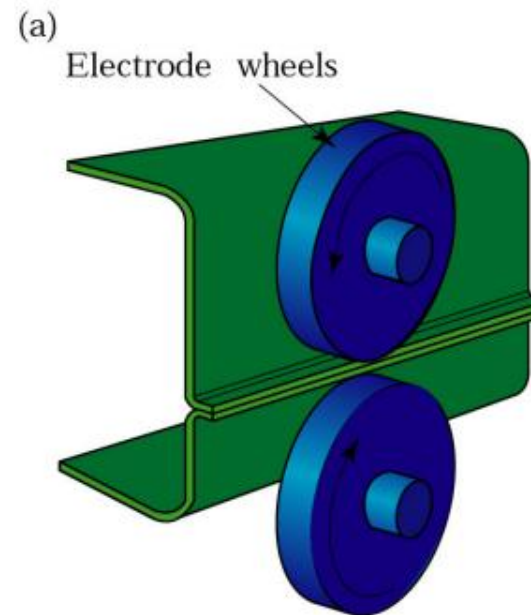


RESISTANCE SPOT WELDING (RSW)



RESISTANCE SEAM WELDING (RSEW)

- A modification of spot welding wherein the electrodes are replaced by rotating wheels or rollers
- Can produce a joint that is liquid tight or gas tight

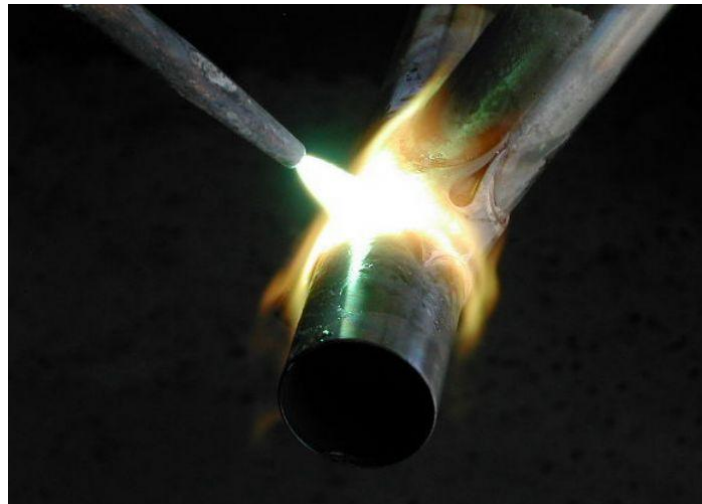
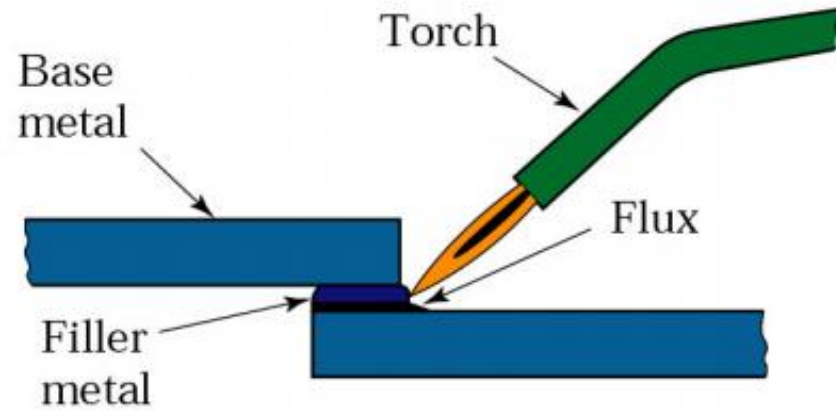


BRAZING AND SOLDERING

BRAZING & SOLDERING

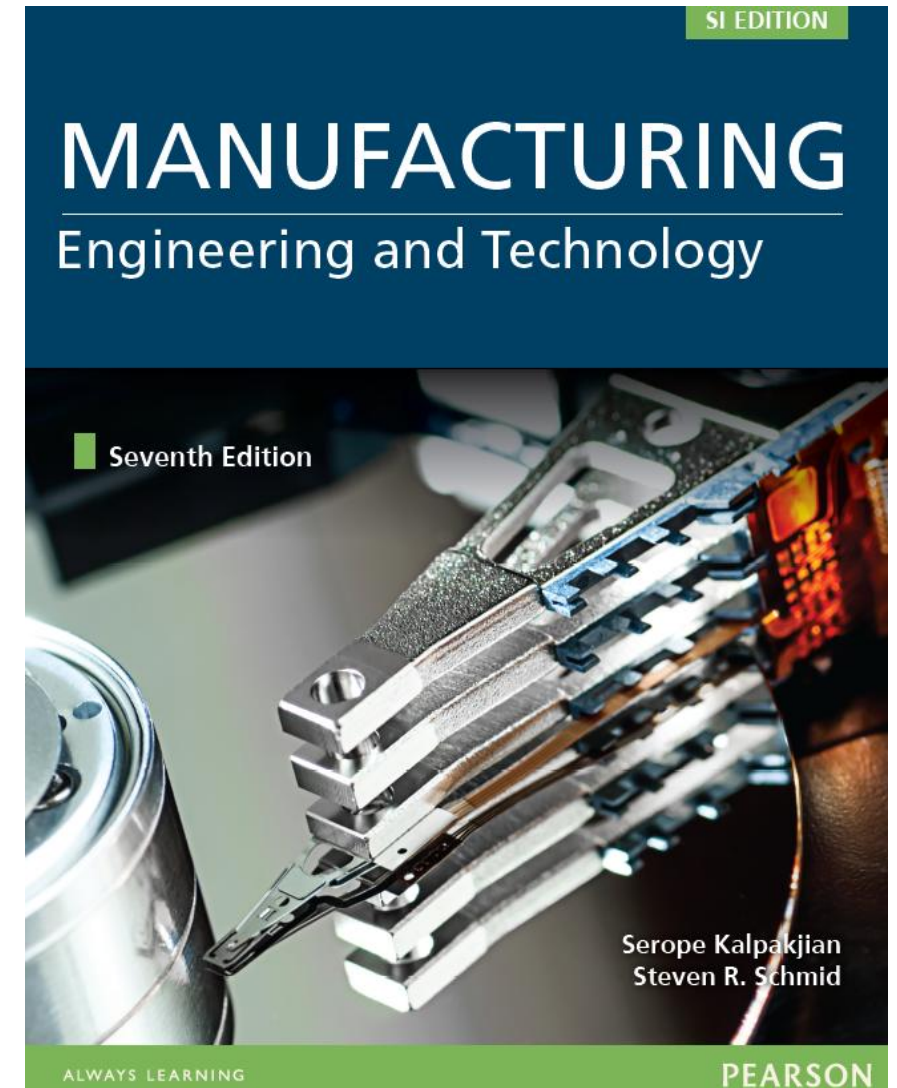
- Brazing
 - A filler metal is placed between the faying surfaces to be joined and the temperature is raised sufficiently melt the filler metal, but not the base metal → *Liquid-solid state* bonding
 - Filler metal melts above 450°C
- Soldering
 - Similar to brazing except filler metal melts below 450°C
- Molten metal is allowed to fill the closely fitting space by *capillary action*

BRAZING & SOLDERING



FURTHER READING

- Serope Kalpakjian and Steven R. Schmid, *Manufacturing Engineering & Technology*, 7th edition, Pearson Education, 2014.
 - Part IV Joining Processes and Equipment





THE END